

SQL Performance & Indexing Optimization Handbook

Deep dive into B-Trees, Execution Plans, and Query Tuning

1. How B-Tree Indexes Work

B-Trees keep data sorted with $O(\log N)$ search, insertion, and deletion.

Leftmost Prefix Rule: Index on (A, B, C) can satisfy queries on A, (A,B), and (A,B,C).

It CANNOT satisfy queries on B or C alone without scanning the whole index.

Avoid functions on indexed columns: WHERE UPPER(email) = ... disables direct index search.

2. ACID & Transaction Isolation Levels

Read Uncommitted: Lowest isolation; allows dirty reads.

Read Committed: Default in PostgreSQL; prevents dirty reads, allows non-repeatable reads.

Repeatable Read: Prevents non-repeatable reads using MVCC snapshots.

Serializable: Highest isolation; strictly ordered execution via predicate locks / SSI.